Department of Computer Science and Engineering

Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: III Semester B.E. ECE ‘A’

Course Code & Title: CS3353 & C Programming and Data Structures

Name of the Faculty member (s): Mrs. K.Jeyakarthika, AP/CSE

Date of Implementation: 05.12.2023

Innovative Practice Description

- **Unit / Topic:** Unit IV / Hashing
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO10
- **Activity Chosen:** Collaborative Learning
- **Justification:**
  - Collaborative learning is an educational approach to teaching and learning that involves groups of students working together to solve a problem.
  - By collaborative learning students will get in-depth knowledge of the particular topics and also making the students to accomplish tasks together is to help students learn the complexities of solving a problem and promote deeper learning through doing.
  - It helps the students to make the concepts more interesting and set them apart from the regular syllabus.
  - Time Allotted for the Activity: 20 minutes

❖ **Details of the Implementation:**
  - Instructor explained the particular concepts in classroom within 30 minutes.
  - Based on the discussion and after clarifying the doubts raised by the students, the teacher asked the students to make the group on own.
  - Problem statements on “Hashing- Collision resolving techniques” were given to the students and insisted to the start the preparation.
  - All the students actively practiced and solved the problem.
  - One of the team members explain the answers to other groups.
  - Finally, faculty member consolidated the information that was discussed in this activity.
  - This assisted the students in recalling the topic taught on that day, generating new ideas about the topic, and answering questions about the topic with ease.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO12
CO5	3	3	2	2	1	1	2	1

PO / PSO mapped:

Innovative Practice	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO12
	3	3	2	2	1	1	2	1
	Able to apply basic knowledge of engineering in different sorting and sorting algorithms.	Able to identify relevant sorting, searching problem for the given applications	Able to interpret the complex engineering problems with the help of the algorithms	Research-based knowledge will help to identify the suitable sorting and searching algorithms.	Usage of Visualgo.net tool for performing various operations using sorting and searching algorithms.	Students can be able to perform individually the appropriate searching and sorting operations for the given application.	Students will be able to explain and present different searching and sorting algorithms.	Student will become aware of the need for lifelong learning and the continued upgrading of technical knowledge using the concept of searching and sorting techniques.

- **Images / Screenshot of the practice:**

- **Reflective Critique:**
  - ❖ **Feedback of practice from students and other stakeholders:**
    - Students expressed that the activity helped to identify the understanding level of the concept.
    - Students also told that the activity motivates them to be attentive in the class for asking doubts further.
  - ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
    - Students were able to explain the concepts clearly, and the activity helped the instructor analyze, evaluate, and enhance their own learning.
    - The students can gain more understanding of the specific topic from this activity.
  - ❖ **Challenges faced in implementation:**
    - Most of the students actively participated except few students. They are not involved to share their understanding level and raising the doubts.

References:

1. https://www.edsys.in/what-is-peer-teaching/
2. https://www.opencolleges.edu.au/informed/features/peer-teaching/
3. https://tilt.colostate.edu/TipsAndGuides/Tip/180
4. https://www.gdrc.org/kmgmt/c-learn/
5. https://www.edutopia.org/topic/collaborative-learning
6. https://www.ritrjpm.ac.in/images/computer-science/2022-2023/Unit_1_collaborative%20learning.pdf